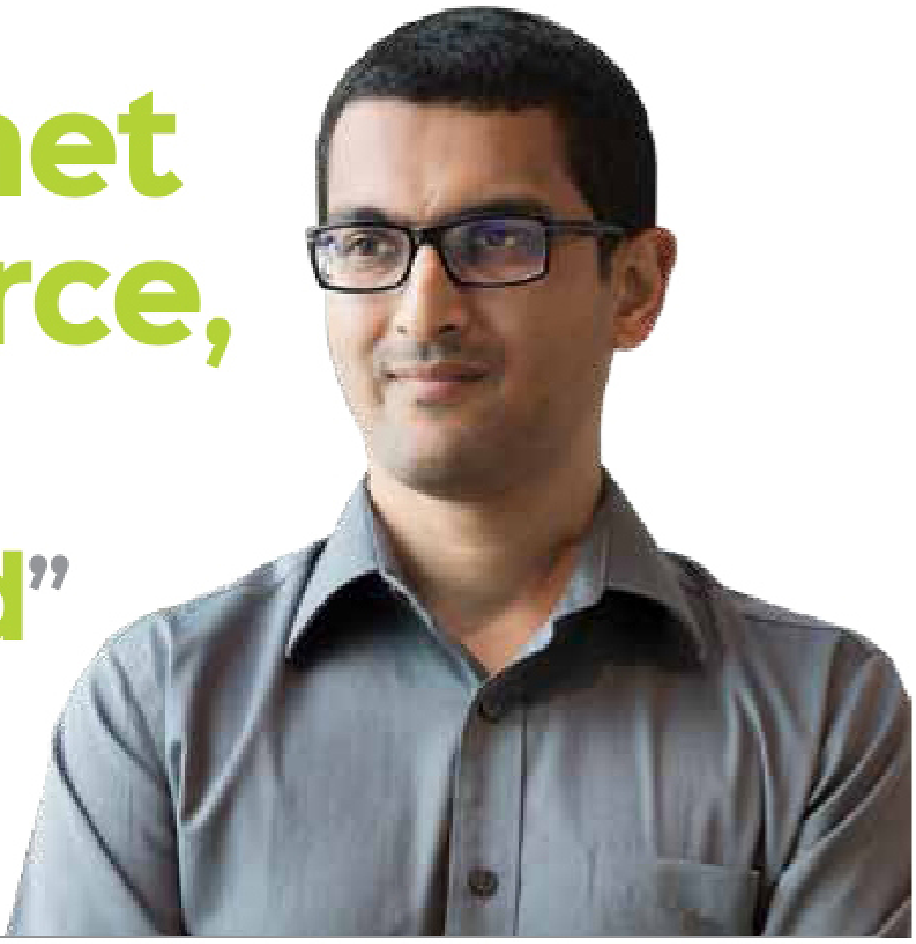
"With the Internet and open source, the world is your playground"



Viral B. Shah »

Viral B. Shah is the co-creator of the Julia programming language along with Alan Edelman, Jeff Bezanson, and Stefan Karpinski. The four of them, together with Keno Fischer and Deepak Vinchhi, founded the company Julia Computing. Shah was also involved in the Aadhaar project. He has co-authored the book 'Rebooting India' with Nandan Nilekani, co-founder and non-executive chairman of Infosys, that talks about their experience of working on and executing the project. He was awarded the J.H. Wilkinson Prize for Numerical Software for his work on the Julia Programming Language in 2019.

In an exclusive chat with *OSFY's* **Sreejani Bhattacharyya,** he talks about his open source journey, growth, challenges, and gives some advice to youngsters starting out.

Shah recalls that he was first introduced to open source software through the *PC Quest* magazine, which used to ship Linux CDs with it. He learnt about Linux, GNU tools like GCC and Emacs, as well as the writings of Richard Stallman from that magazine. He was impressed by what Debian was doing and soon became a Debian developer. After that, he got busy with his PhD and then started the Julia project with his co-creators. Shah has a PhD in computer science from the University of California, Santa Barbara.

He says, "My contributions have always been in the space of scientific software since I have believed in the power of science and technology to solve some of the world's most challenging problems. Scientific software plays a special role because it helps us understand, model, design, and predict, making it a powerful tool to solve the challenges we face, whether it is COVID-19, climate change, clean energy, conservation, and even social inequality and justice!"

Shah's early contributions were as a Debian developer helping maintain the mosix kernel patches. Since then, his major open source projects have been the Julia programming language and Circuitscape, which is modelling software for ecological conservation.

He adds, "Due to my love for scientific software, a lot of the challenges encountered tend to be mathematical and scientific. In my work with Circuitscape, we faced a number of challenges with devising the right mathematical solvers for large landscapes that span entire continents. So we were not just developing open source software, but also developing new methods in the field of landscape ecology."

Designing the Julia programming language and making it open source

Shah recalls that when the work on Julia was started, they faced the two language problem. They solved it